



# **SUPERSTORE CASE STUDY: SQL INTEGRATION & POWER BI VISUALIZATION**

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# PROJECT OVERVIEW

- **Dataset:** Superstore (9.994 rows, 21 columns)
- **Objective:** Analyze sales performance, profit trends, and regional distribution
- **Tools:** MySQL for data processing, Power BI for visualization

| Result Grid    Filter Rows: <input type="text"/>   Export:    Wrap Cell Content:    Fetch rows:  <input type="text"/> |        |                |            |            |                |             |                 |           |               |                 |            |             |        |          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------|------------|------------|----------------|-------------|-----------------|-----------|---------------|-----------------|------------|-------------|--------|----------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Row ID | Order ID       | Order Date | Ship Date  | Ship Mode      | Customer ID | Customer Name   | Segment   | Country       | City            | State      | Postal Code | Region | Product  |
| ▶                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1      | CA-2016-152156 | 11/8/2016  | 11/11/2016 | Second Class   | CG-12520    | Claire Gute     | Consumer  | United States | Henderson       | Kentucky   | 42420       | South  | FUR-BO-  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2      | CA-2016-152156 | 11/8/2016  | 11/11/2016 | Second Class   | CG-12520    | Claire Gute     | Consumer  | United States | Henderson       | Kentucky   | 42420       | South  | FUR-CH-  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3      | CA-2016-138688 | 6/12/2016  | 6/16/2016  | Second Class   | DV-13045    | Darrin Van Huff | Corporate | United States | Los Angeles     | California | 90036       | West   | OFF-LA-  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 4      | US-2015-108966 | 10/11/2015 | 10/18/2015 | Standard Class | SO-20335    | Sean O'Donnell  | Consumer  | United States | Fort Lauderdale | Florida    | 33311       | South  | FUR-TA-  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 5      | US-2015-108966 | 10/11/2015 | 10/18/2015 | Standard Class | SO-20335    | Sean O'Donnell  | Consumer  | United States | Fort Lauderdale | Florida    | 33311       | South  | OFF-ST-  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 6      | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosina Hoffman | Consumer  | United States | Los Angeles     | California | 90032       | West   | FUR-FU-  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 7      | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosina Hoffman | Consumer  | United States | Los Angeles     | California | 90032       | West   | OFF-AR-  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 8      | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosina Hoffman | Consumer  | United States | Los Angeles     | California | 90032       | West   | TEC-PH-  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 9      | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosina Hoffman | Consumer  | United States | Los Angeles     | California | 90032       | West   | OFF-BI-1 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 10     | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosina Hoffman | Consumer  | United States | Los Angeles     | California | 90032       | West   | OFF-AP-  |

# BEFORE CLEANING

- **Data Types:** All as TEXT
- **Issues:** Missing & inconsistent values
- **Next Step:** Convert & standardize

Invalid  
entry here



| Result Grid |                |
|-------------|----------------|
|             | Sales          |
| ▶           | Executive Red" |
|             | Black"         |
|             | Black          |
|             | Light Blue"    |
|             | Gregg Ruled"   |
|             | 200 Messages"  |
|             | Card"          |

Result Grid |  Filter Rows:  | Export:  | Wrap Cell (

|   | Field         | Type | Null | Key | Default | Extra |
|---|---------------|------|------|-----|---------|-------|
| + | Row ID        | text | YES  |     | NULL    |       |
|   | Order ID      | text | YES  |     | NULL    |       |
|   | Ship Mode     | text | YES  |     | NULL    |       |
|   | Customer ID   | text | YES  |     | NULL    |       |
|   | Customer Name | text | YES  |     | NULL    |       |
|   |               |      |      |     | NULL    |       |

# DATA CLEANING STEPS

- **Convert Columns:**  
Numeric & Date
- **Handle Missing:** Fill or remove
- **Remove Duplicates:**  
**SELECT DISTINCT**
- **Standardize Formats:**  
Text & categories

```
1 CREATE TABLE superstore_clean AS
2 SELECT DISTINCT *
3 FROM superstore;
4
5 • DROP TABLE superstore;
6 • RENAME TABLE superstore_clean TO superstore;
```



|   | Row ID | Order ID       | Ship Mode      | Customer ID | Customer Name   | Segment   | Country       | City            | State      | Postal Code | Region | Product ID      | Category        |
|---|--------|----------------|----------------|-------------|-----------------|-----------|---------------|-----------------|------------|-------------|--------|-----------------|-----------------|
| ▶ | 1      | CA-2016-152156 | Second Class   | CG-12520    | Claire Gute     | Consumer  | United States | Henderson       | Kentucky   | 42420       | South  | FUR-BO-10001798 | Furniture       |
|   | 2      | CA-2016-152156 | Second Class   | CG-12520    | Claire Gute     | Consumer  | United States | Henderson       | Kentucky   | 42420       | South  | FUR-CH-10000454 | Furniture       |
|   | 3      | CA-2016-138688 | Second Class   | DV-13045    | Darrin Van Huff | Corporate | United States | Los Angeles     | California | 90036       | West   | OFF-LA-10000240 | Office Supplies |
|   | 4      | US-2015-108966 | Standard Class | SO-20335    | Sean O'Donnell  | Consumer  | United States | Fort Lauderdale | Florida    | 33311       | South  | FUR-TA-10000577 | Furniture       |

| Sub-Category | Product Name                                      | Sales  | Quantity | Discount | Profit | Order Date | Ship Date  |
|--------------|---------------------------------------------------|--------|----------|----------|--------|------------|------------|
| Bookcases    | Bush Somerset Collection Bookcase                 | 261.96 | 2        | 0.00     | 41.91  | 2016-11-08 | 2016-11-11 |
| Chairs       | Hon Deluxe Fabric Upholstered Stacking Chairs,... | 731.94 | 3        | 0.00     | 219.58 | 2016-11-08 | 2016-11-11 |
| Labels       | Self-Adhesive Address Labels for Typewriters b... | 14.62  | 2        | 0.00     | 6.87   | 2016-06-12 | 2016-06-16 |
| Tables       | Bretford CR4500 Series Slim Rectangular Table     | 957.58 | 5        | 0.45     | 0.00   | 2015-10-11 | 2015-10-18 |

# INITIAL ANALYSIS

Result Grid

  Filter Rows:

|  | Region  | Total_Sales | Total_Profit |
|--|---------|-------------|--------------|
|  | West    | 713471.45   | 128591.44    |
|  | East    | 672194.15   | 139862.26    |
|  | Central | 497800.88   | 95512.71     |
|  | South   | 388983.59   | 73403.81     |

Result Grid





Filter Rows:

|   | Month   | Total_Sales | Total_Profit |
|---|---------|-------------|--------------|
| ▶ | 2014-01 | 14161.35    | 3071.88      |
|   | 2014-02 | 4119.84     | 883.20       |
|   | 2014-03 | 55526.23    | 4857.35      |
|   | 2014-04 | 28139.56    | 4373.66      |
|   | 2014-05 | 23634.66    | 4490.05      |
|   | 2014-06 | 34509.01    | 6273.31      |

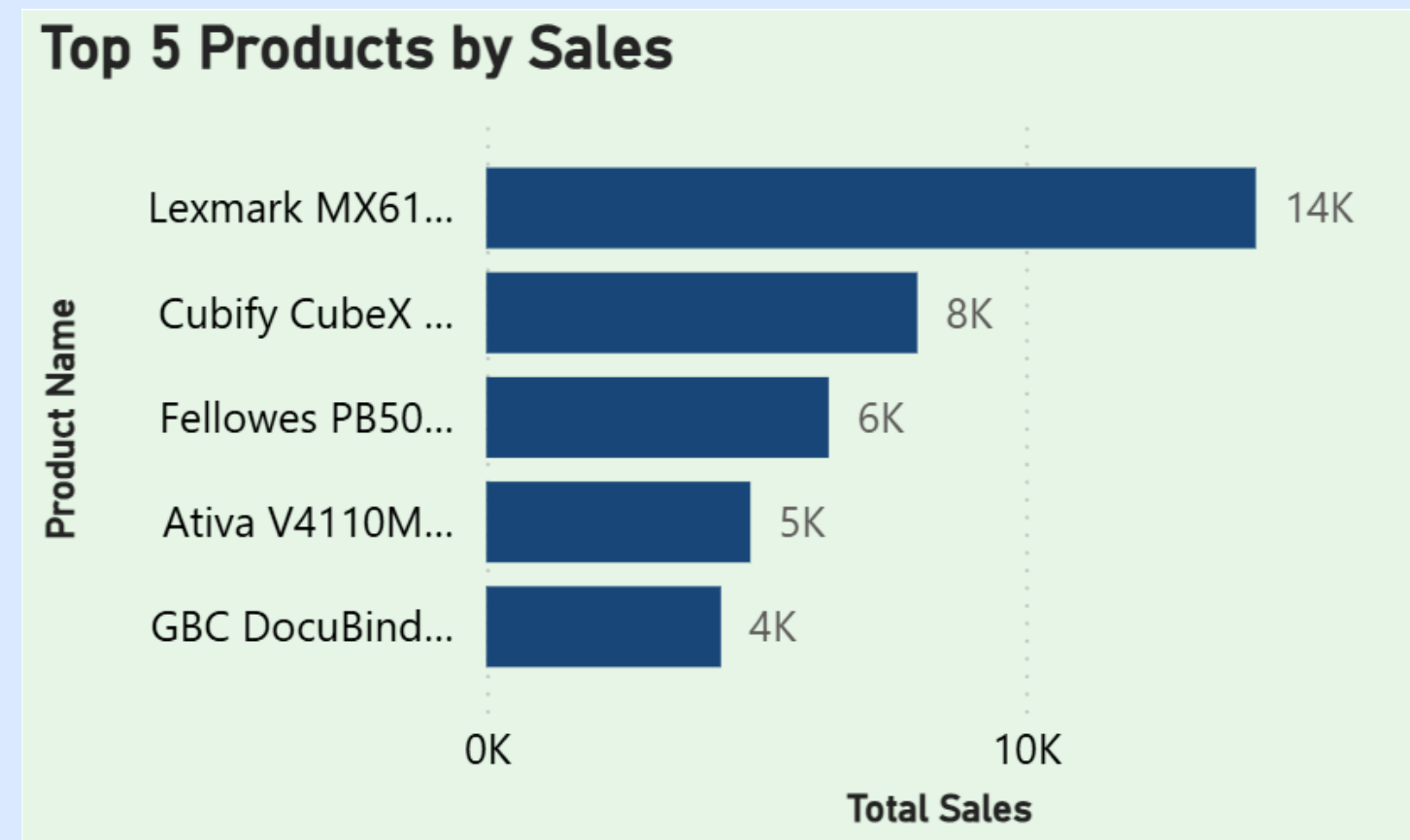
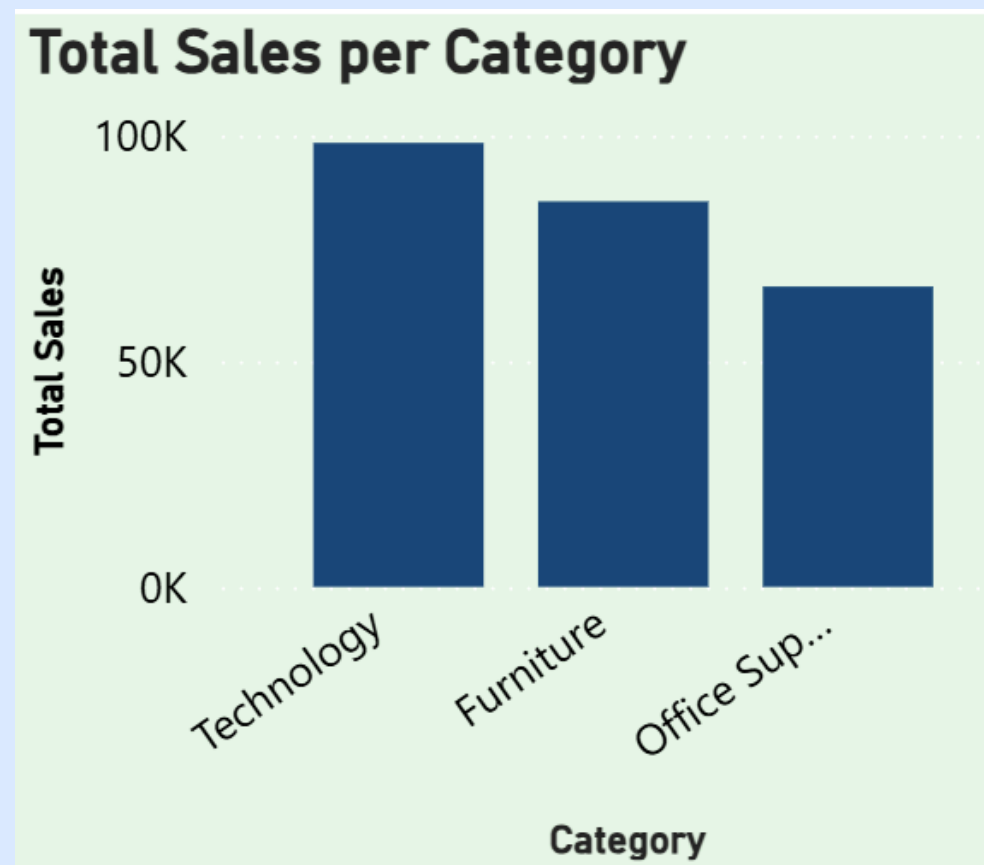
|             |                 |                                                                                     |                                                                                                  |  |
|-------------|-----------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--|
| Result Grid |                 |  |  Filter Rows: |  |
|             | Category        | Total_Sales                                                                         | Total_Profit                                                                                     |  |
| ▶           | Technology      | 835900.14                                                                           | 183967.85                                                                                        |  |
|             | Furniture       | 733047.06                                                                           | 77469.92                                                                                         |  |
|             | Office Supplies | 703502.87                                                                           | 175932.45                                                                                        |  |

| Product Name                                       | Total_Sales |
|----------------------------------------------------|-------------|
| Canon imageCLASS 2200 Advanced Copier              | 61599.83    |
| Fellowes PB500 Electric Punch Plastic Comb Bind... | 27453.38    |
| Cisco TelePresence System EX90 Videoconferen...    | 22638.48    |
| HON 5400 Series Task Chairs for Big and Tall       | 21870.57    |
| GBC DocuBind TL300 Electric Binding System         | 19823.48    |

Cleaned dataset ready to generate insights: sales performance, profit trends, and regional distribution.

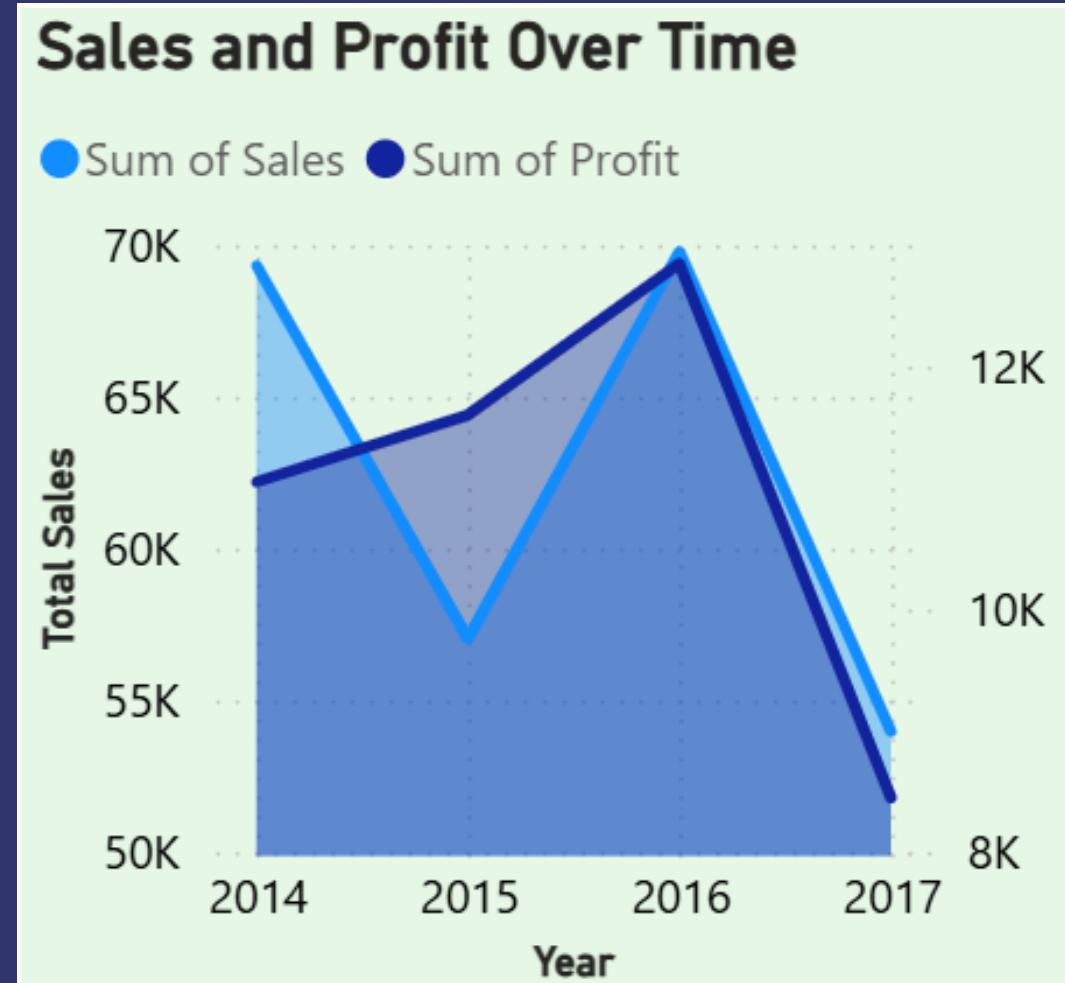
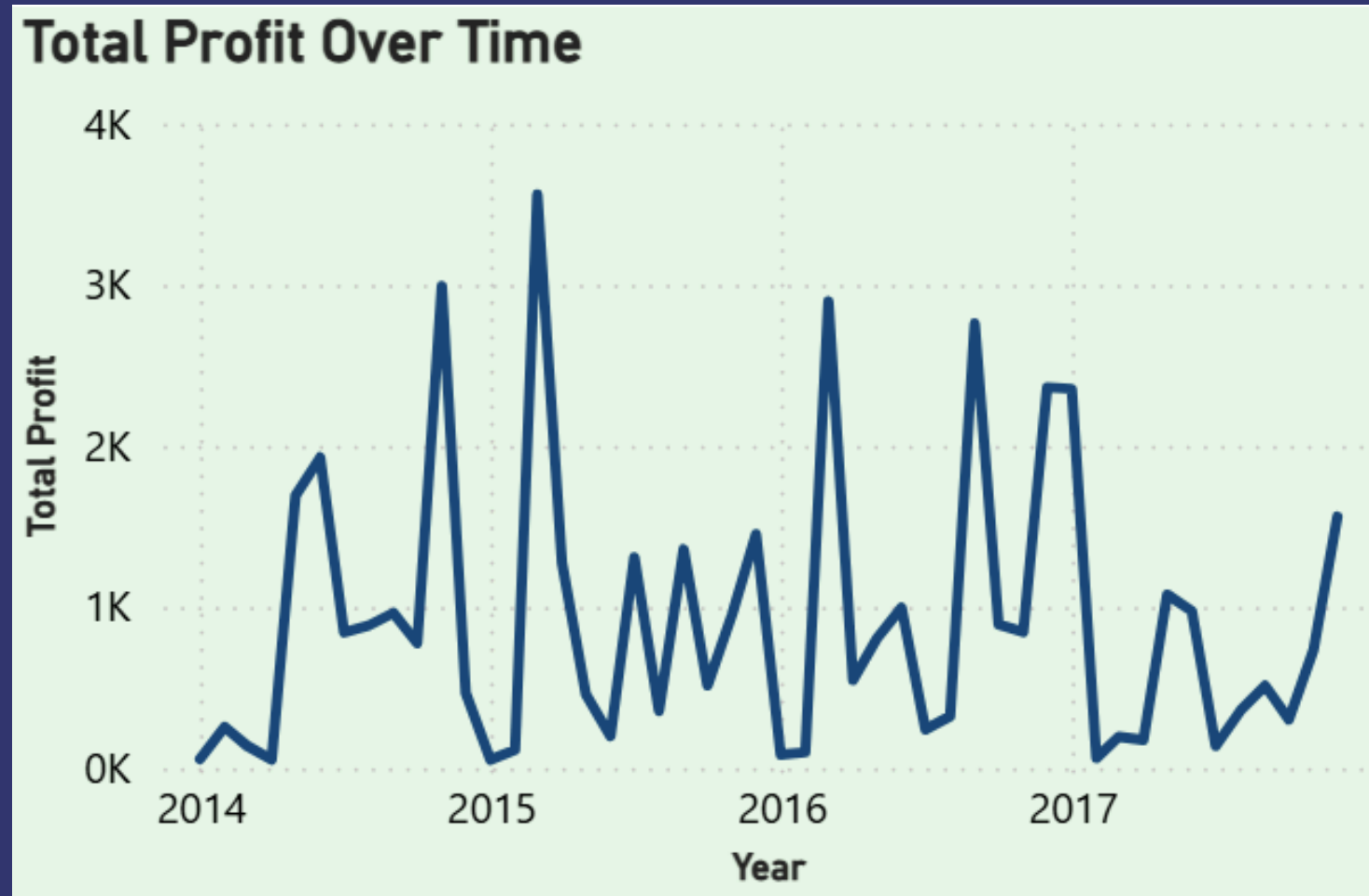


# SALES PERFORMANCE



- **Technology** generated **the highest sales**, followed by Furniture and Office Supplies.
- A few products **dominate sales**, with **Lexmark MX611dhe** leading at ~14K.
- Sales distribution shows clear concentration in top categories and products.

# PROFIT & SALES TRENDS



- **Sales (Area Chart):** Highest in 2016 (~70K), lowest in 2017 (~52K).
- **Profit (Line Chart):** Peaked in 2015, then declined steadily to the lowest in 2017.

# REGIONAL SALES OVERVIEW

## Regional Sales Distribution

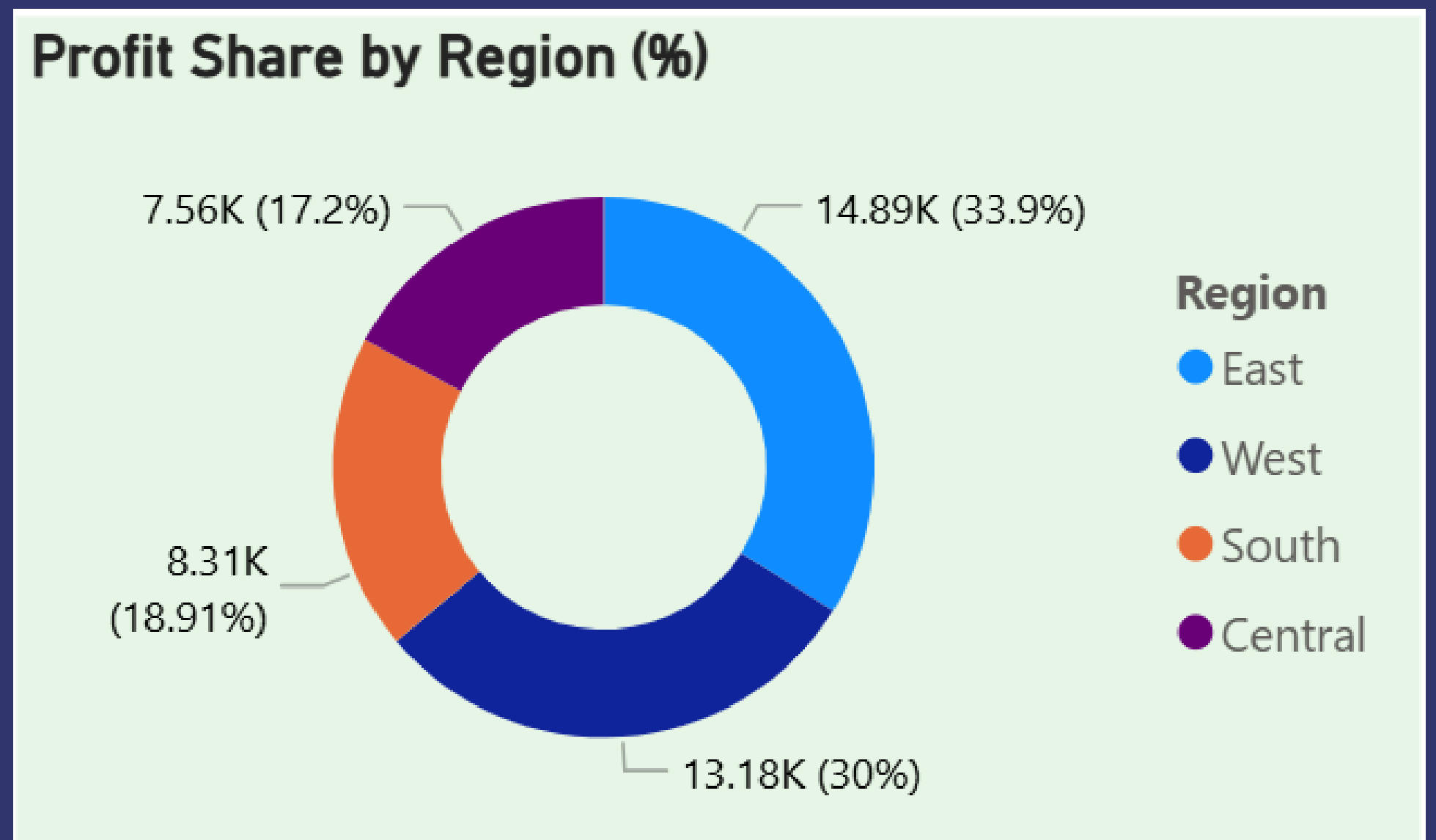


- Sales are **highest** in the **East** and **West** regions
- The **Central** region shows **moderate** sales
- The **South** region contributes the **least** to total sales



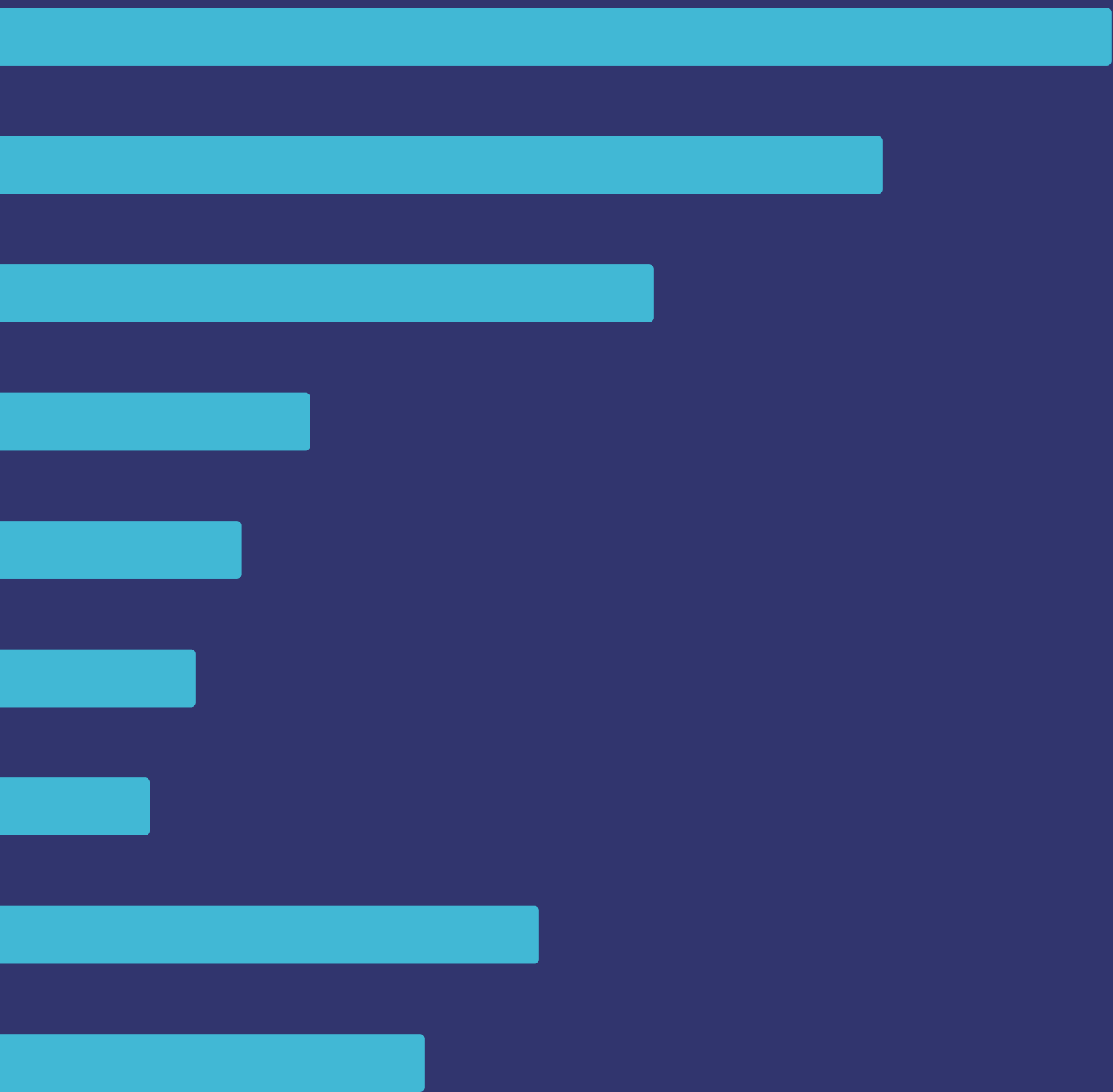
# REGIONAL PROFIT CONTRIBUTION (%)

- **East** Region contributes the **largest** share (**33.9%**)
- **West** Region follows with a significant share (**30%**)
- **South** Region accounts for **18.9%**
- **Central** Region shows the **smallest** contribution (**17.2%**)



# KEY BUSINESS INSIGHTS

- **Technology** is the leading category in total sales.
- **Profit trends** show fluctuations over time, highlighting seasonality.
- **East** and **West** regions are the **strongest** contributors to both sales and profit.
- **South** region shows the **lowest sales**, but contributes **more profit than Central** due to higher margins.
- **Central** region consistently **underperforms** in profitability.



**Thank You!**